

Summer Supplementary Examination – 2023

Semester: V**Duration: 3 Hr.**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Determine which of the signals are periodic or non-periodic and compute their fundamental period. a) $x(n) = \cos(0.2\pi n)$ b) $x(n) = n^2$	(L1/CO1)	6
B) Write any 6 properties of Discrete Time Fourier Transform.	(L1/CO2)	6
C) Find and sketch even and odd parts of the given signals a) $u(n)$ b) $x(n) = [4, -2, \underset{\uparrow}{4}, -6]$	(L1/CO1)	6
Q.2 Solve Any Two of the following.		12
A) Find the Fourier transform of given signals. Also plot magnitude and phase. a) $x(n) = \delta(n)$ b) $x(n) = a^n u(n)$	(L2/CO2)	6
B) a) Compute the 4-point DFT of $x(n) = (-1)^n \dots 0 \leq n \leq 3$ using matrix method. b) Perform linear convolution using circular convolution of given signals. $x(n) = \{2, 5, 0, 4\}$, $h(n) = \{4, 1, 3\}$	(L3/CO1)	6
C) Given $x(n) = \{1, 0, -1, 0, 1, 0, -1, 0\}$. Find X(k) using DIT-FFT algorithm.	(L3/CO4)	6
Q. 3 Solve Any Two of the following.		12
A) Find the inverse z-transform using partial fraction method for given signal. $X(Z) = \frac{1-0.5Z^{-1}}{1-0.75Z^{-1}}$	(L2/CO2)	6

- B)** Use initial value theorem to find the initial value of the signals. **(L3/CO1)** **6**
- a)** $X(Z) = \frac{2+Z^{-1}}{(1-Z^{-1})(1+0.5Z^{-1})}$
- b)** $X(Z) = \frac{1-3Z^{-1}}{(1-0.1Z^{-1})(1+0.6Z^{-1})}$
- C)** Consider the following difference equation $y[n]=x[n]-0.5y[n-1]$. Find the Z-transform of this difference equation assuming zero initial conditions. **(L3/CO4)** **6**

Q.4 Solve Any Two of the following.

12

- A)** Compare and contrast Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters in terms of their characteristics, advantages, disadvantages **(L2/CO2)** **6**
- B)** Given an analog low-pass filter with the transfer function. transform it into a digital filter using the Impulse Invariant Method with a sampling period of $T=0.1$ seconds. **(L3/CO1)** **6**

$$H_a(S) = \frac{1}{(S+1)}$$

- C)** For transfer function $H(S)$ find $H(Z)$ using Bilinear transformation method. Assume $T=1$. **(L2/CO2)** **6**

$$H(S) = \frac{1}{(S^2 + \sqrt{2}S + 1)}$$

Q. 5 Solve Any Two of the following.

12

- A)** A digital Butterworth filter has to be designed using Bilinear transformation. The filter specifications are as follows: **(L3/CO2)** **6**

$$\begin{aligned} 0.9 \leq |H(e^{j\omega})| \leq 1 & \quad 0 \leq \omega \leq 0.5\pi \\ |H(e^{j\omega})| \leq 0.2 & \quad 0.75\pi \leq \omega \leq \pi \end{aligned}$$

Find the filter order N and cut-off frequency Ω_c

- B)** Find the filter order N and cut-off frequency Ω_c for an IIR low pass Butterworth filter using Bilinear transformation. The filter specifications are as follows: **(L2/CO4)** **6**

$$\begin{aligned} \text{Passband: } 0.8 \leq |H(e^{j\omega})| \leq 1 & \quad |\omega| \leq 0.2\pi \\ \text{Stopband: } |H(e^{j\omega})| \leq 0.2 & \quad 0.6\pi \leq |\omega| \leq \pi \end{aligned}$$

Assume $T=1$ second.

- C)** Consider the following LTI system with system function and draw the direct form I and direct form II structure **(L2/CO2)** **6**

a) $H(Z) = 1 - \frac{1}{3}Z^{-1} + \frac{1}{6}Z^{-2} + Z^{-3}$

b) $H(Z) = \frac{1+2Z^{-1}-Z^{-2}}{1+Z^{-1}-Z^{-2}}$

***** End *****